

Open Science

Today's agenda:

- Final presentation reminder
- Code and Data sharing
- Coding time in class (optional)

Final Presentations

- 12:30-2:30 p.m. on Tuesday, December 9th
- Remember to have your slides added to the Google Slides file beforehand
- Practice your timing to make sure you're under the 5 minutes

What is Open Science?

Open Science: “transparent and accessible knowledge that is shared and developed through collaborative networks”

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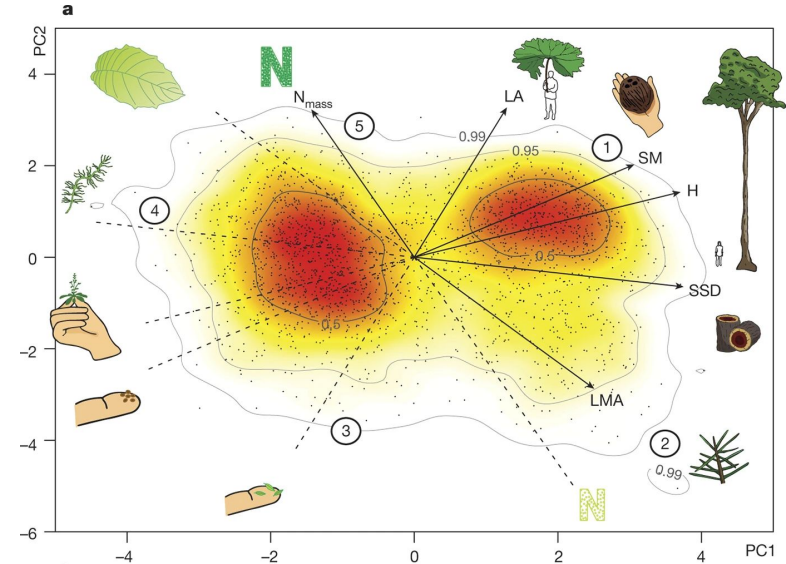
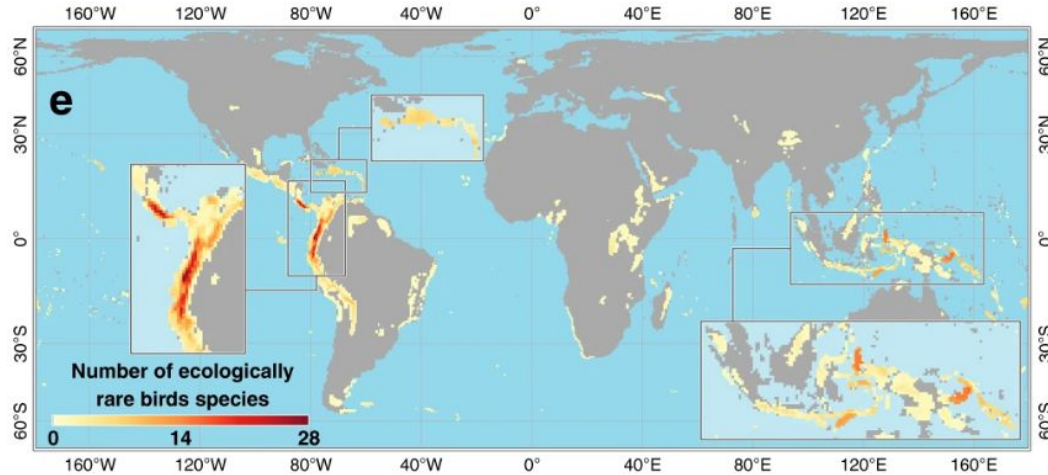


What are Open Data?

- **Available and Accessible:** the data must be *available as a whole* and at no more than a reasonable reproduction cost, preferably by downloading over the internet. The data must also be available in a *convenient and modifiable form*.
- **Reusable and Redistributable:** the data must be provided under terms that *permit re-use and redistribution* including the intermixing with other datasets.
- **Universal Participation:** *everyone must be able to use*, re-use and redistribute - there should be no discrimination against fields of endeavour or against persons or groups.



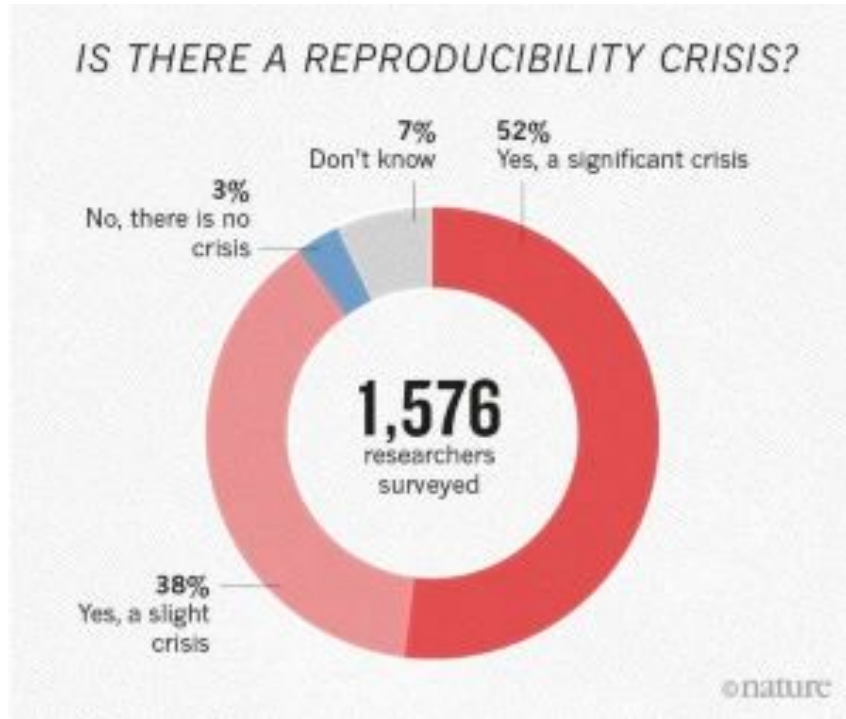
Global Questions Need Global Participation



Global spectrum of plant form and function



Open Data Enhance Reproducibility



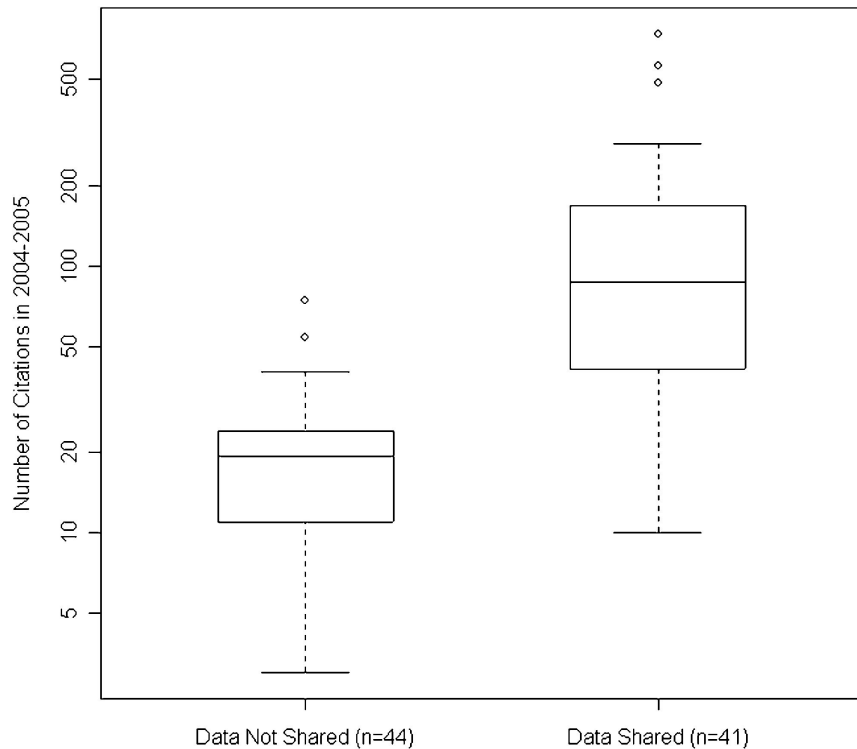


Ethical Responsibilities

- **Publicly funded data are public property**
- **Transparency is a key part of science**
- **Open Data increase inclusion**
- **Open Data save money!**

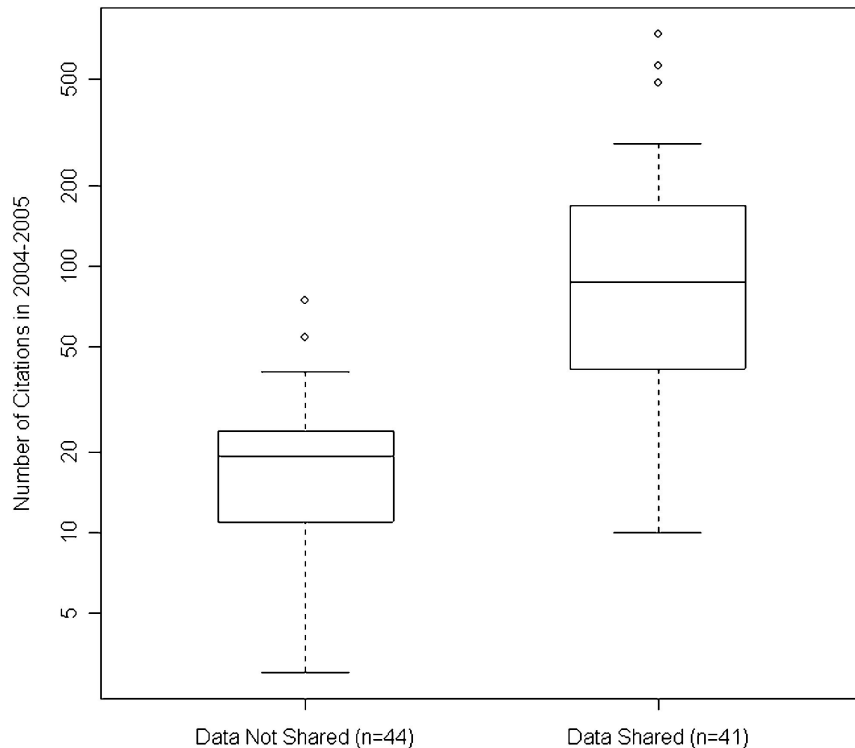


Open Data Improve Citations





Open Data Improve Citations and Publications



www.nature.com/scientificdata

scientific **data**

OPEN DATA DESCRIPTOR

Check for updates

Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard

Vigdis Vandvik^{1,2,5}, Aud H. Halbritter^{1,2}, Inge H. J. Althuizen^{2,3}, Casper T. Christiansen⁴, Jonathan J. Henn⁵, Ingibjörg Svala Jónsdóttir⁶, Kari Klanderud⁷, Marc Macias-Fauria⁸, Yadvinder Malhi⁹, Brian Salvin Maitner⁹, Sean Michaletz¹⁰, Ruben E. Roos⁷, Richard J. Telford¹, Polly Bass¹¹, Katrín Björnsdóttir⁴, Lucely Lucero Vilca Bustamante¹², Adam Chmurszynski¹³, Shuli Chen¹, Siri Vatsø Haugum^{1,4}, Julia Kemppinen¹¹, Kai Lepley¹⁴, Yaoqi Li¹⁵, Mary Linabury¹⁶, Ilaine Silveira Matos¹⁷, Barbara M. Neto-Bradley¹⁸, Molly Ng¹⁹, Pekka Niittynen¹³, Silje Østman¹, Karolína Pánková²⁰, Nina Roth²¹, Matis Castorena¹, Marcus Spiegel¹⁴, Eleanor Thomson⁸, Alexander Sæle Vågenes¹ & Brian J. Enquist^{9,10}



How to share data

Range of options



How to share data

Range of options

The screenshot shows the Zenodo interface for a dataset. The header is blue with the Zenodo logo, a search bar, and links for 'Communities' and 'My dashboard'. There are 'Log in' and 'Sign up' buttons. Below the header, a grey bar features the Dryad logo and name. The main content area has a light blue background. It displays the publication date 'Published March 11, 2019 | Version v1' and buttons for 'Dataset' and 'Open'. The title of the dataset is 'Data from: Relationships between plant traits, soil properties and carbon fluxes differ between monocultures and mixed communities in temperate grassland'. Below the title, the authors are listed: 'De Long, Jonathan R.¹; Jackson, Benjamin G.²; Wilkinson, Anna¹; Pritchard, William J.¹; Oakley, Simon³; Mason, Kelly E.³; Stephan, Jörg G.⁴; Ostle, Nicholas J.⁵; Johnson, David¹; Baggs, Elizabeth M.²; Bardgett, Richard D.¹'. A 'Show affiliations' button is located below the authors. On the right side, there are statistics: '17 VIEWS' and '9 DOWNLOADS', with a 'Show more details' link. Below this is a 'Versions' section showing 'Version v1' with the DOI '10.5061/dryad.gh41n3j' and the date 'Mar 11, 2019'.

zenodo Search records... Communities My dashboard Log in Sign up

Dryad

Published March 11, 2019 | Version v1 Dataset Open

Data from: Relationships between plant traits, soil properties and carbon fluxes differ between monocultures and mixed communities in temperate grassland

De Long, Jonathan R.¹; Jackson, Benjamin G.²; Wilkinson, Anna¹; Pritchard, William J.¹; Oakley, Simon³; Mason, Kelly E.³; Stephan, Jörg G.⁴; Ostle, Nicholas J.⁵; Johnson, David¹; Baggs, Elizabeth M.²; Bardgett, Richard D.¹

Show affiliations

17 VIEWS 9 DOWNLOADS Show more details

Versions

Version v1 10.5061/dryad.gh41n3j Mar 11, 2019



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Dryad

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figshare Browse Search on figshare... Q Log in Sign up

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Growth.form	Growth.form.2	AccSpeciesName	HV (cm/cm2)	Al/As (m2/cm2)	HV (10*4, m2/m2)	K (g/m2)	P (g/m2)	N (g/m2)	K (kg/g)	P (g/kg)	N (g/kg)	LMA (g/m
2	liana	liana	Acacia Megaladena										
3	liana	liana	Bauhinia aurea										
4	liana	liana	Bauhinia yunnanensis										
5	liana	liana	Byttneria aspera										
6	liana	liana	Combretum latifolium		1.09	1.54						19.93	
7	liana	liana	Ficus sagittata										
8	liana	liana	Freycinetia macleodii										
9	liana	liana	Malvula repens										
10	liana	liana	Phoradendron scandens										

full_met_analysis_data_8_Feb.csv (27.81 kB)

Processed extended meta-analysis of hydraulic functional traits

[Cite](#) [Download \(27.81 kB\)](#) [Share](#) [Embed](#) [+ Collect](#)

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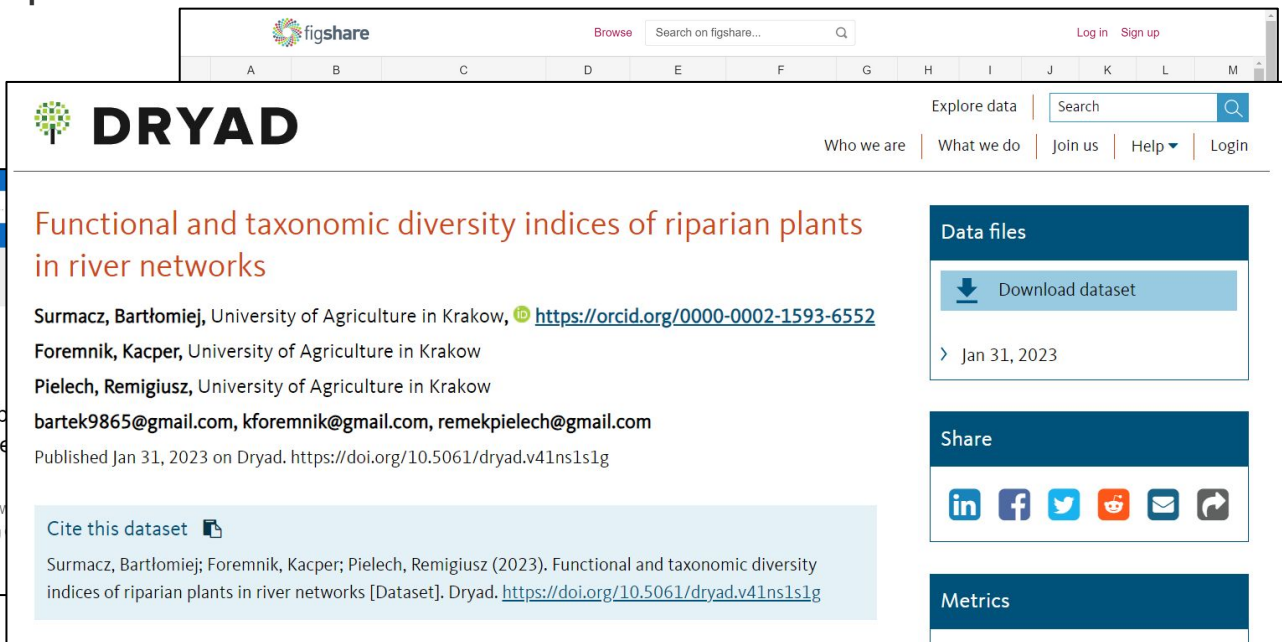
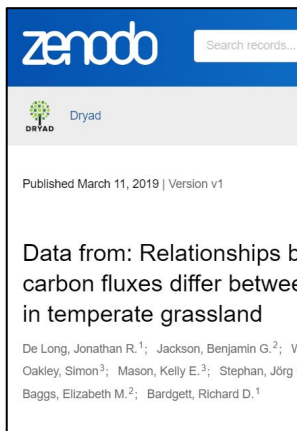
Version v1 Mar 11, 2019

10.5061/dryad.gh41n3j



How to share data

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How to share data

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Published March 11, 2019 | Version v1

Data from: Relationships between carbon fluxes differ between temperate grassland

De Long, Jonathan R.¹; Jackson, Benjamin G.²; V
Oakley, Simon³; Mason, Kelly E.³; Stephan, Jörg
Baggs, Elizabeth M.²; Bardgett, Richard D.¹

DRYAD

Functional and taxonomic diversity indices of riparian plants in river networks

Surmacz, Bartłomiej, University of Agriculture
Foremnik, Kacper, University of Agriculture
Pielech, Remigiusz, University of Agriculture
bartek9865@gmail.com, kforemnik@gmail.com
Published Jan 31, 2023 on Dryad. <https://doi.org/10.5061/dryad.v41ns1slg>

Cite this dataset

Surmacz, Bartłomiej; Foremnik, Kacper; Pielech, Remigiusz (2023). Functional and taxonomic diversity indices of riparian plants in river networks [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v41ns1slg>

OSFHOME

Data for: Are sub-alpine species' seedling emergence and establishment in the alpine limited by climate or biotic interactions? (Dahle et. al 2023)

Contributors: Ingrid Dahle, Vigdis Vandvik, Ragnhild Gya, Joachim Töpper
Date created: 2023-09-29 03:10 AM | Last Updated: 2023-09-29 03:20 AM
Identifier: DOI 10.17605/OSF.IO/79VGT
Category: Data
Description: Data for the paper "Are sub-alpine species' seedling emergence and establishment in the alpine limited by climate or biotic interactions?"

501.8KB

Metrics



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in river netw

Surmacz, Bartłomiej
Foremnik, Kacper, U
Pielech, Remigiusz,
bartek9865@gmail.

Published Jan 31, 2023 on Dryad. <https://doi.org/10.5061/dryad.v41ns1slg>

Cite this dataset

Surmacz, Bartłomiej; Foremnik, Kacper; Pielech, Remigiusz (2023). Functional and taxonomic diversity indices of riparian plants in river networks [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v41ns1slg>

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AdamWilsonLab / emma_envdata

Code Pull requests Actions Projects Wiki Security Insights Settings

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Nov 1, 2022
bmaintrner
v1.0
c04a8c0
Compare

v1.0 Latest
This release contains the code as of 2022-11-01, and is working as expected.

Assets 2

Aug 16, 2022
bmaintrner
raw_ndvi_date...
3cb3c1d
Compare

raw_ndvi_dates_viirs
Data release

Assets 528

Identifier: DOI 10.17605/OSF.IO/79VGT

Category: Data

Description: Data for the paper "Are sub-alpine species' seedling emergence and establishment in the alpine limited by climate or biotic interactions?"

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Dryad

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Support

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bartek9865@gmail.

Published Jan 31, 2023 on Dryad. <https://doi.org/10.5061/dryad.v41ns1slg>

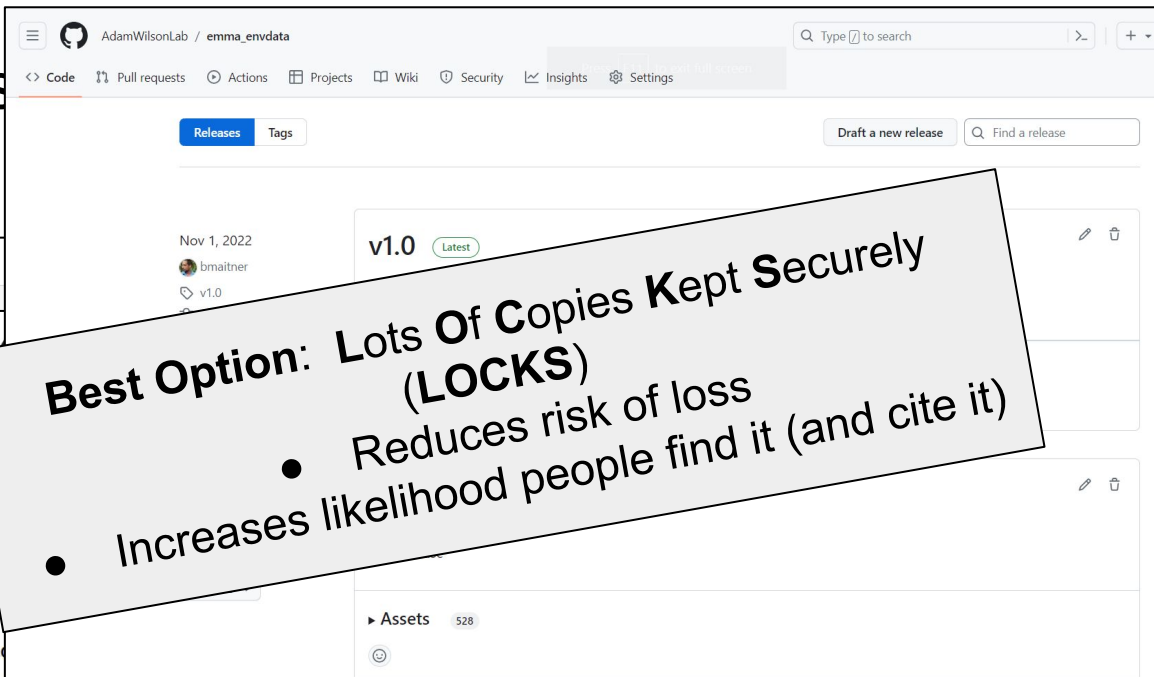
Cite this dataset

Surmacz, Bartłomiej; Foremnik, Kacper; Pielech, Remigiusz (2023). Functional and taxonomic diversity indices of riparian plants in river networks [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v41ns1slg>

Metrics

Best Option: Lots Of Copies Kept Securely (LOCKS)

- Reduces risk of loss
- Increases likelihood people find it (and cite it)



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What is Open Source?

- **Free Redistribution** The license shall not restrict any party from selling or giving away the software as a component of an aggregate software distribution containing programs from several different sources. The license ***shall not require a royalty or other fee*** for such sale.
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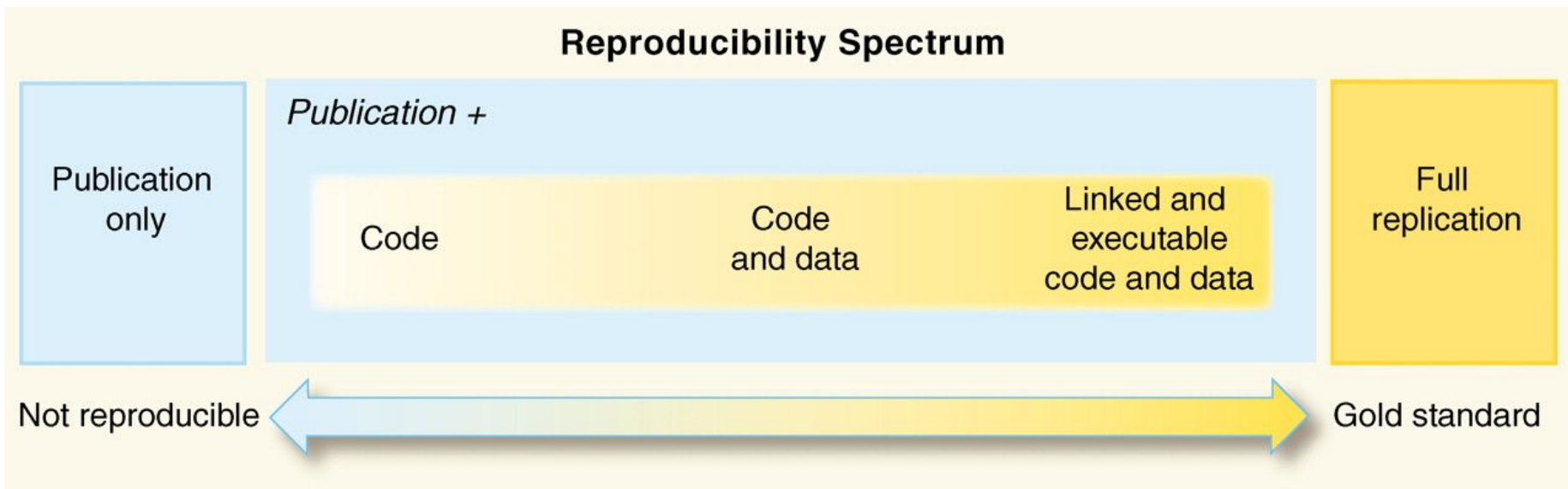


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- **No Discrimination Against Fields of Endeavor** The license must not restrict anyone from making use of the program in a specific field of endeavor. For example, it may not restrict the program from being used in a business, or from being used for genetic research.
- **Code that is free to use, both in terms of costs and restrictions**



Reproducibility and Transparency

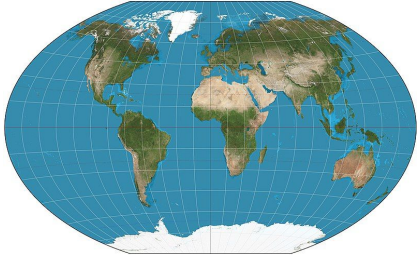




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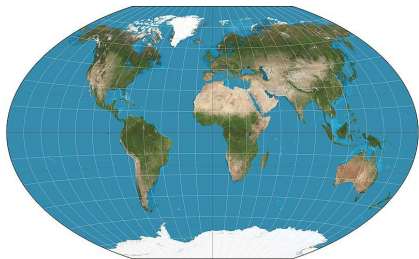


Open Source Reduces Wasted Efforts





Open Source Reduces Wasted Efforts

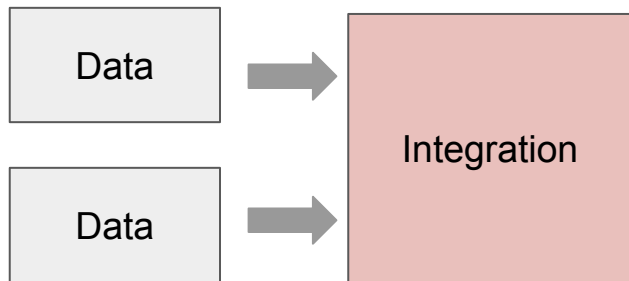
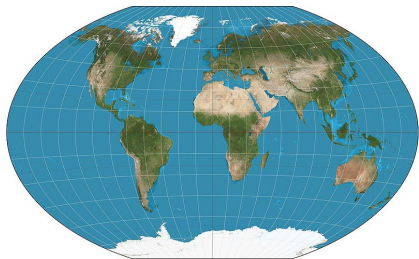


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Data



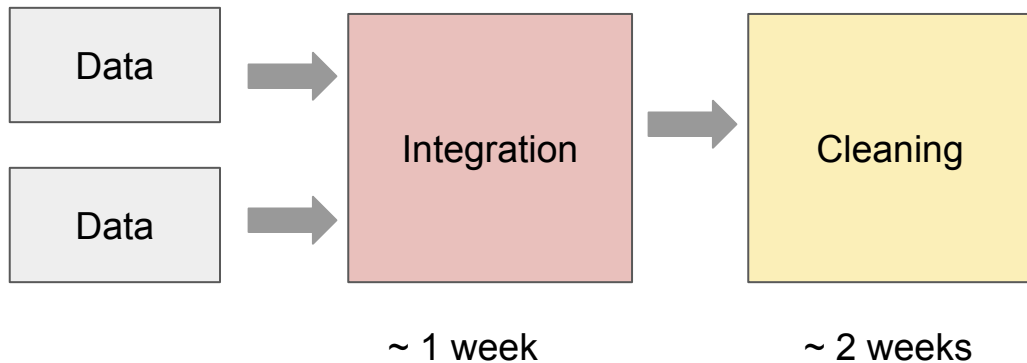
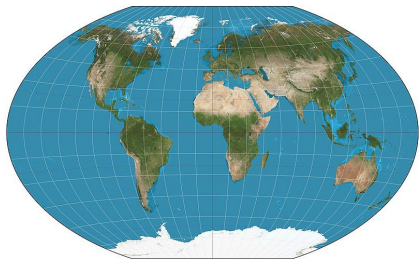
Open Source Reduces Wasted Efforts



~ 1 week

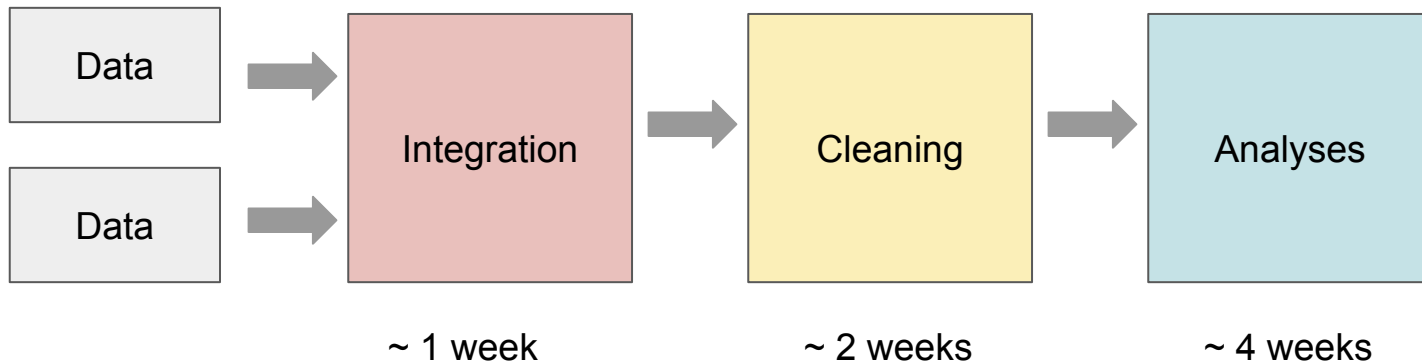
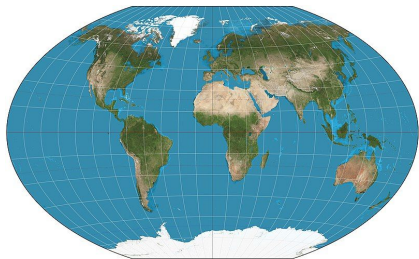


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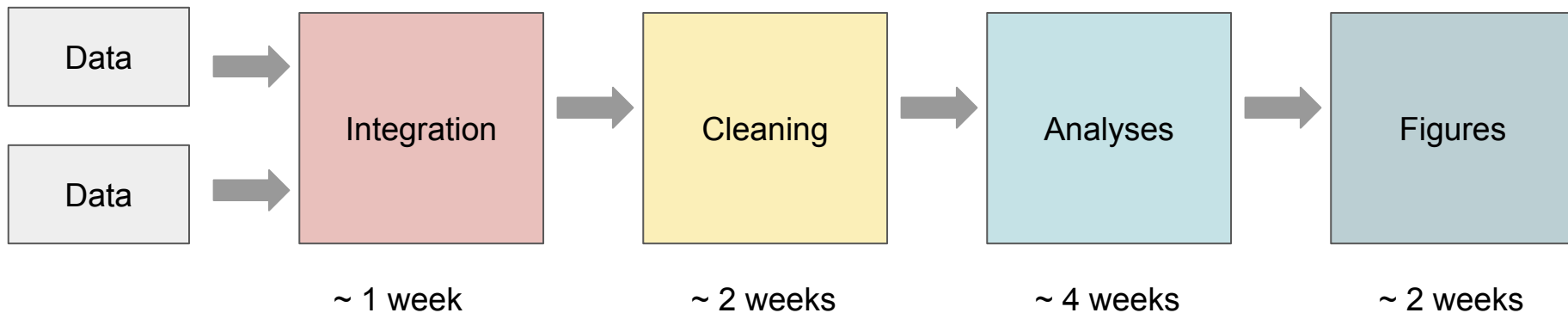
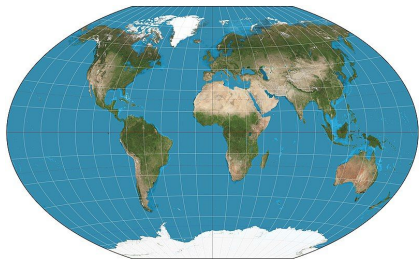


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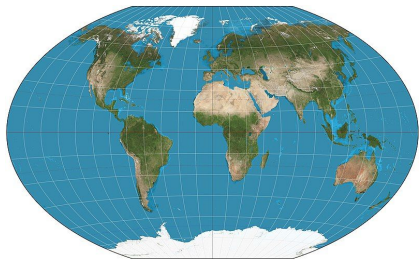


Open Source Reduces Wasted Efforts





Open Source Reduces Wasted Efforts



~ 10 minutes

Publish
Code

Data



Integration



Cleaning



Analyses



Figures

~ 1 week

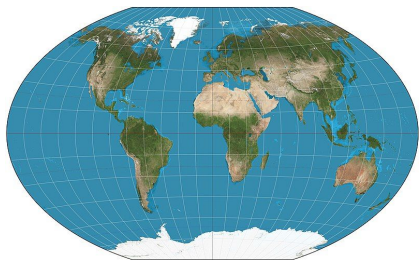
~ 2 weeks

~ 4 weeks

~ 2 weeks



Open Source Reduces Wasted Efforts



~ 10 minutes

Publish
Code



Data



Data



Integration



Cleaning



Analyses



Figures

~ 1 week

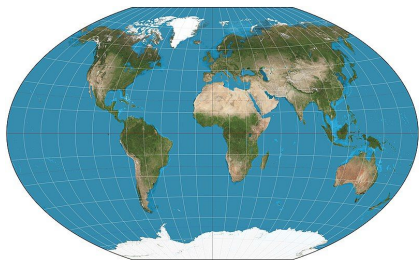
~ 2 weeks

~ 4 weeks

~ 2 weeks

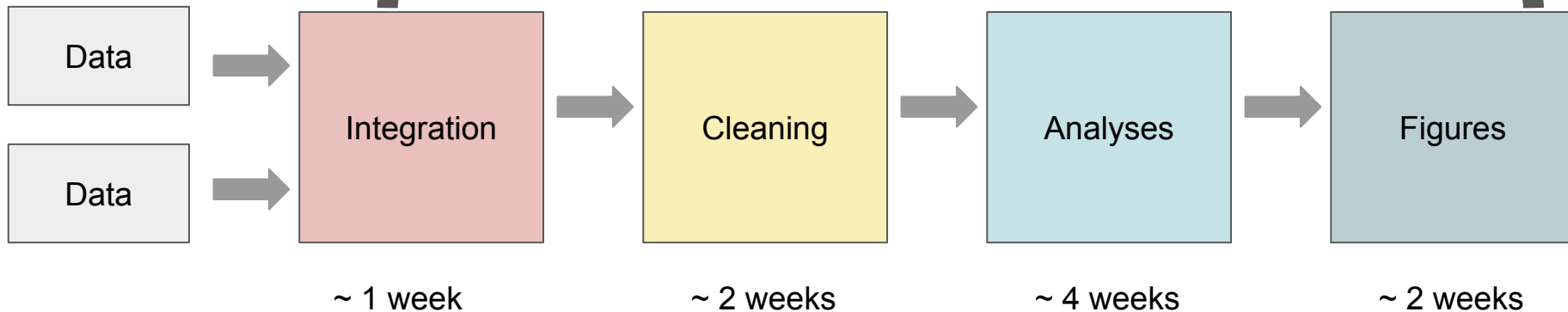


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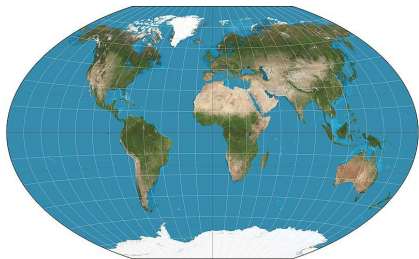
~ 10 minutes

Publish
Code





Open Source Reduces Wasted Efforts



~ 10 minutes

Publish
Code



Data



Data



Integrate

~ 1 week

I saved people months of work!
(and got a few good citations)

~ 2 weeks

~ 4 weeks

Figures

~ 2 weeks



Open Source Reduces Barriers



Open Source Reduces Barriers



ArcGIS Pro All Extensions Bundle

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Get MATLAB



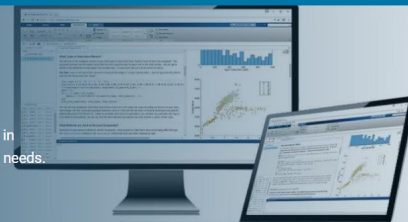
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intended use

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- ☐ Startups
- ☐ Academic
- ☐ Student
- ☐ Home

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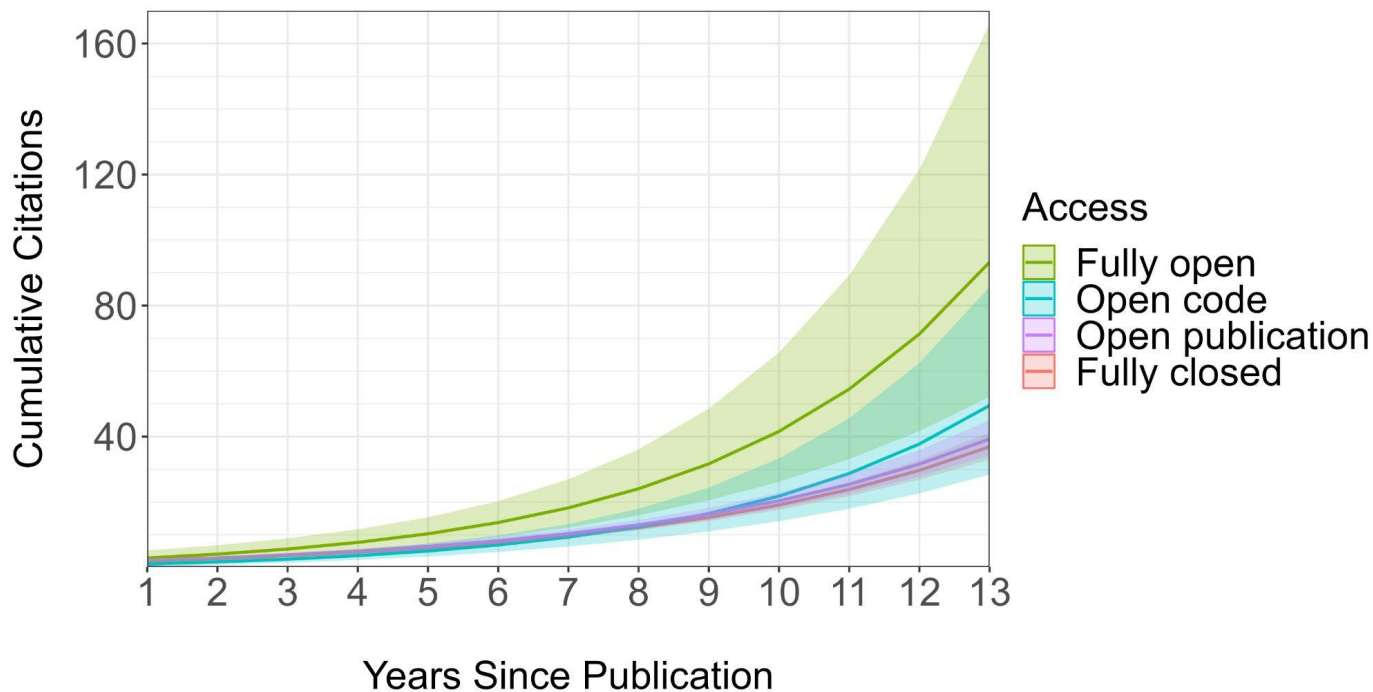
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Code Sharing Increases Citations



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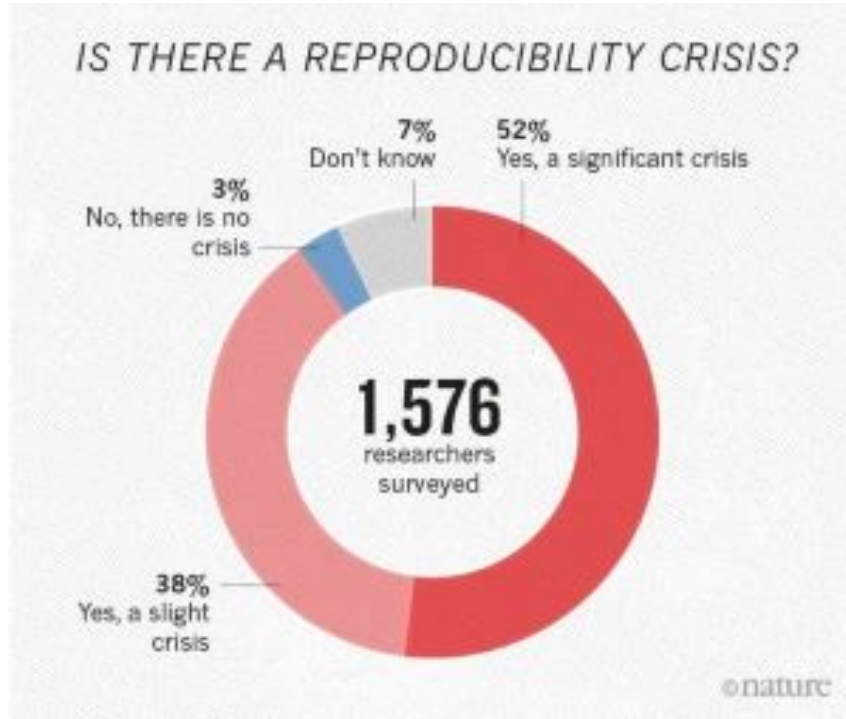


What are Open Methods?

- Making methods underlying research open and freely accessible
- **Examples:**
 - Laboratory protocols
 - Data collection protocols
 - Metadata standards
 - Materials
 - Reagents
 - Code (Overlap with Open Source)



Open Methods Increase Transparency





Shared Methods Increase Confidence

CSIRO PUBLISHING

Australian Journal of Botany, 2013, **61**, 167–234
<http://dx.doi.org/10.1071/BT12225>

New handbook for standardised measurement of plant functional traits worldwide

N. Pérez-Harguindeguy^{A,Y}, S. Díaz^A, E. Garnier^B, S. Lavorel^C, H. Poorter^D, P. Jaureguiberry^A, M. S. Bret-Harte^E, W. K. Cornwell^F, J. M. Craine^G, D. E. Gurvich^A, C. Urcelay^A, E. J. Veneklaas^H, P. B. Reich^I, L. Poorter^J, I. J. Wright^K, P. Ray^L, L. Enrico^A, J. G. Pausas^M, A. C. de Vos^F, N. Buchmann^N, G. Funes^A, F. Quétier^{A,C}, J. G. Hodgson^O, K. Thompson^P, H. D. Morgan^Q, H. ter Steege^R, M. G. A. van der Heijden^S, L. Sack^T, B. Blonder^U, P. Poschold^N, M. V. Vaieretti^A, G. Conti^A, A. C. Staver^W, S. Aquino^X and J. H. C. Cornelissen^F

Received: 31 May 2019 | Accepted: 28 October 2019

DOI: 10.1111/2041-210X.13331

RESEARCH ARTICLE

Methods in Ecology and Evolution 

The handbook for standardized field and laboratory measurements in terrestrial climate change experiments and observational studies (ClimEx)

Aud H. Halbritter¹  | Hans J. De Boeck²  | Amy E. Eycott^{3,4}  | Sabine Reinsch⁵  |
David A. Robinson⁵  | Sara Vicca²  | Bernd Berauer⁶  | Casper T. Christiansen⁷  |
Marc Estiarte^{8,9}  | José M. Grünzweig¹⁰ | Ragnhild Gya¹ | Karin Hansen^{11,12} |
Anke Jentsch⁶  | Hanna Lee⁷  | Sune Linder¹³  | John Marshall¹⁴ |
Josep Peñuelas^{8,9}  | Inger Kappel Schmidt¹⁵  | Ellen Stuart-Haëntjens¹⁶  |
Peter Wilfahrt⁶  | the ClimMani Working Group* | Vigdis Vandvik¹ 



Shared Methods Increase Compatibility

- Data Integration is a major time sink
- Using shared metadata standards and methods increases interoperability

REVIEW

Methods in Ecology and Evolution



Towards an ecological trait-data standard

Florian D. Schneider¹  | David Fichtmueller²  | Martin M. Gossner³  |
Anton Güntsch²  | Malte Jochum^{4,5,6}  | Birgitta König-Ries⁷  |
Gaëtane Le Provost⁸  | Peter Manning⁸  | Andreas Ostrowski⁷  |
Caterina Penone⁴  | Nadja K. Simons^{9,10} 

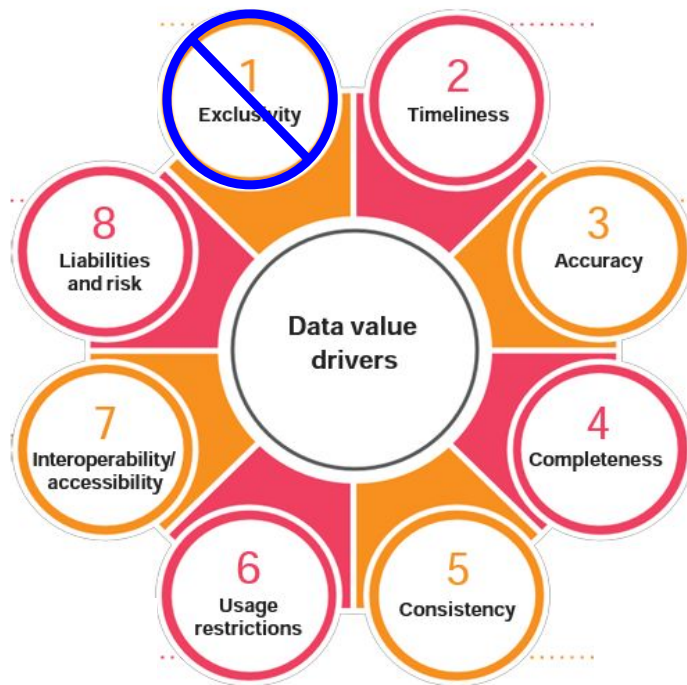


Open Methods Increase Data Value



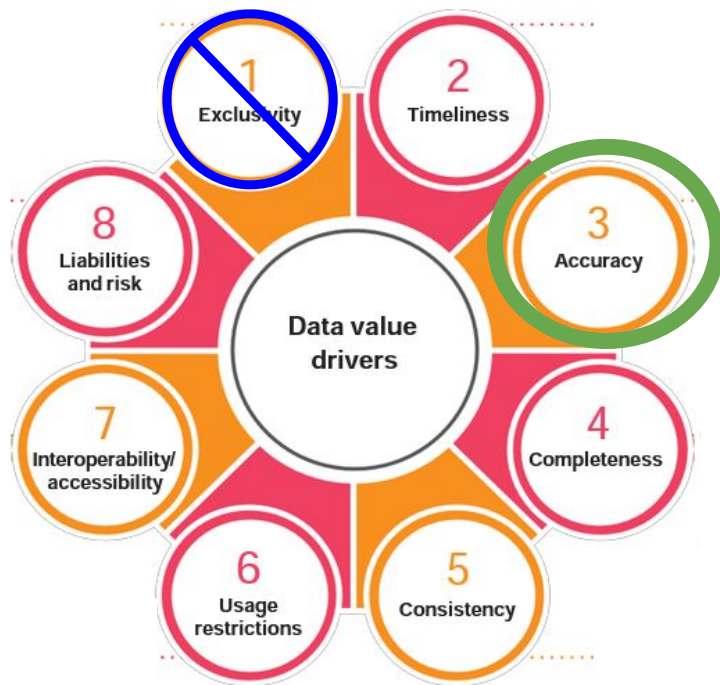


Open Methods Increase Data Value



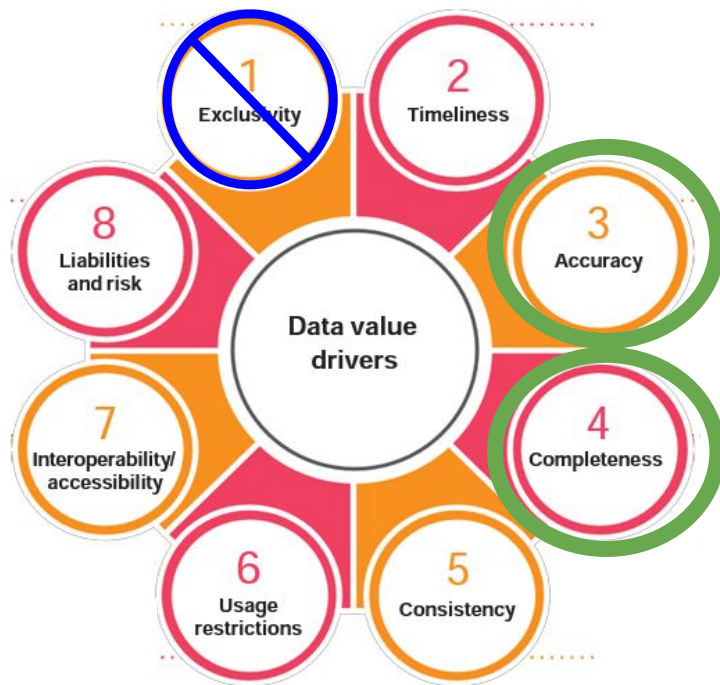


Open Methods Increase Data Value



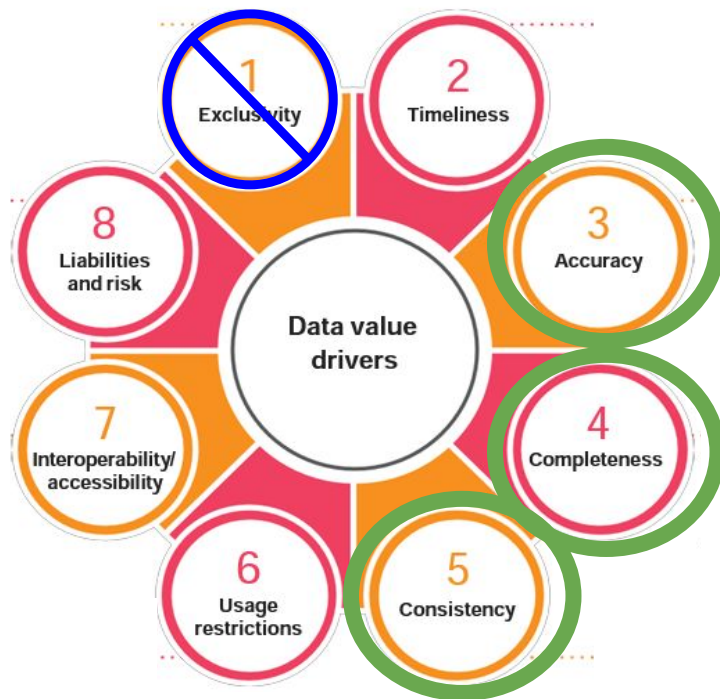


Open Methods Increase Data Value



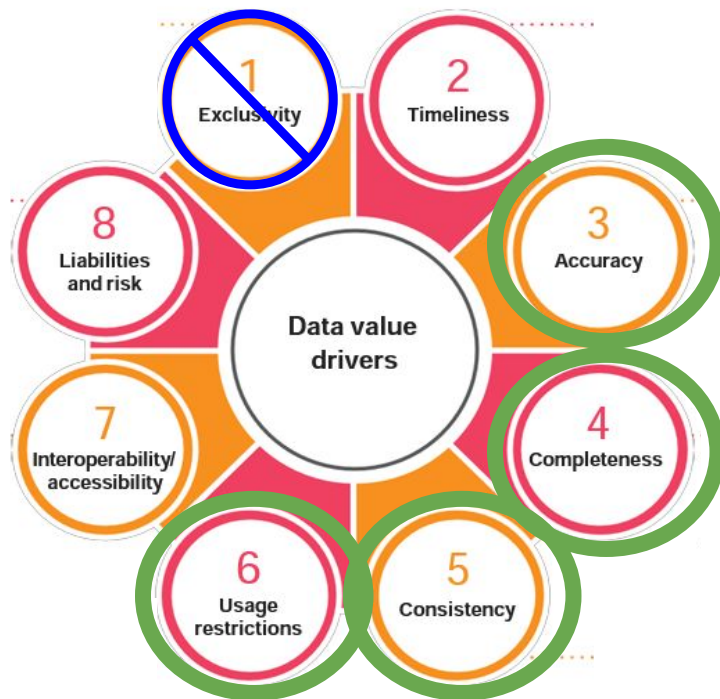


Open Methods Increase Data Value



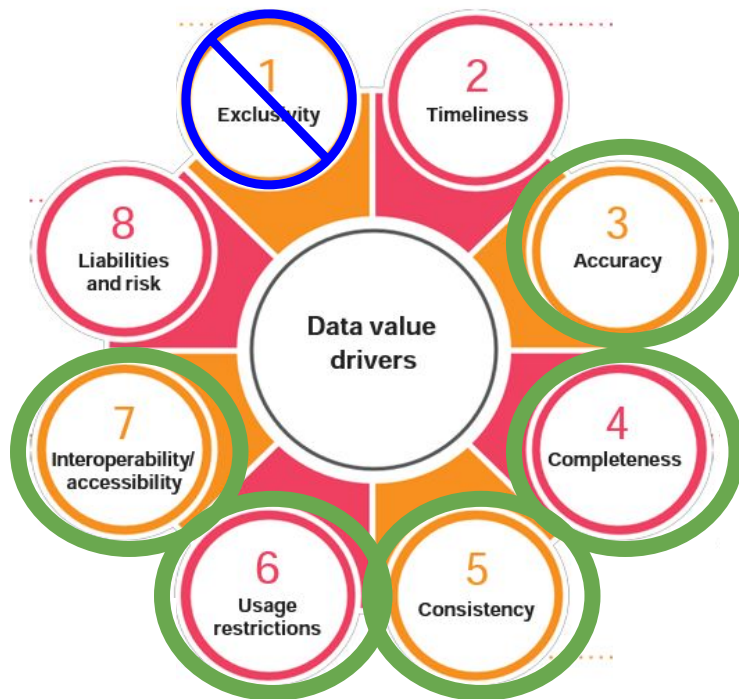


Open Methods Increase Data Value



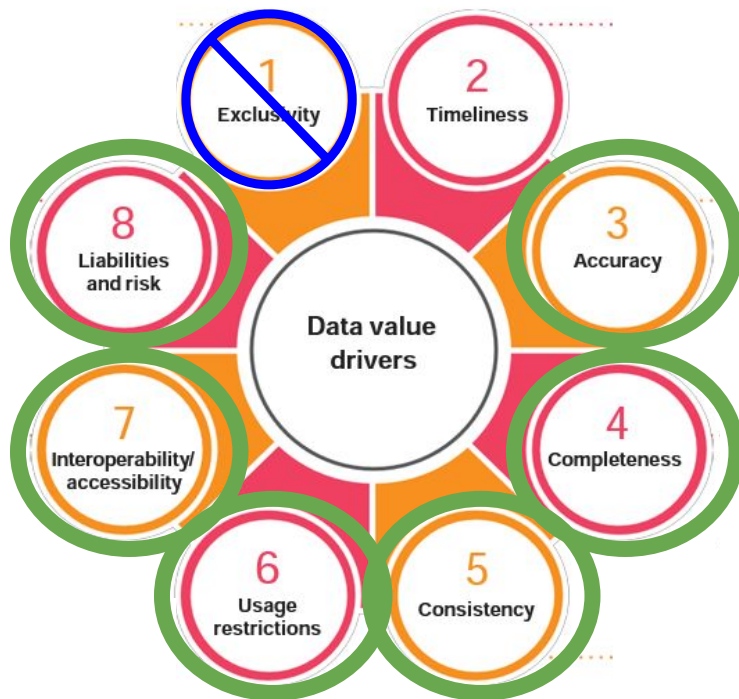


Open Methods Increase Data Value



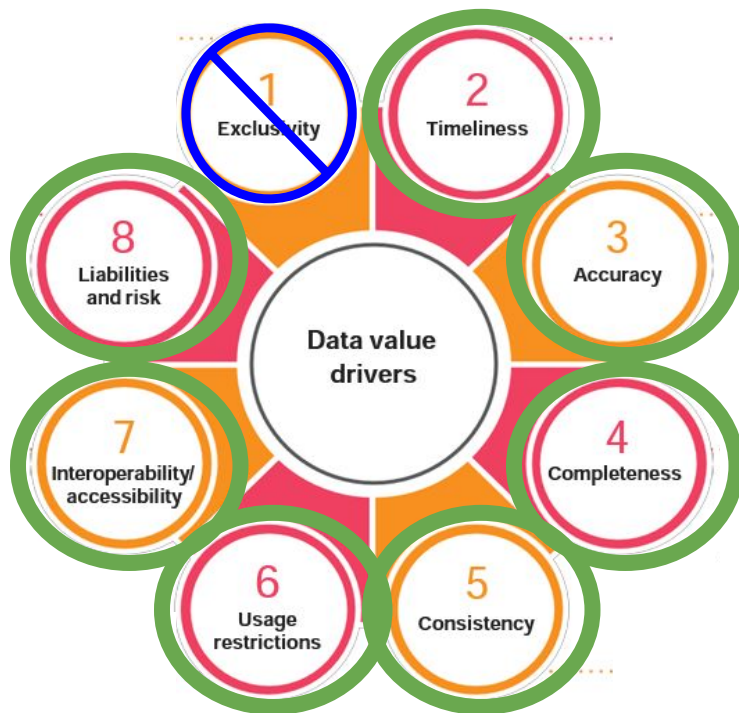


Open Methods Increase Data Value





Open Methods Increase Data Value

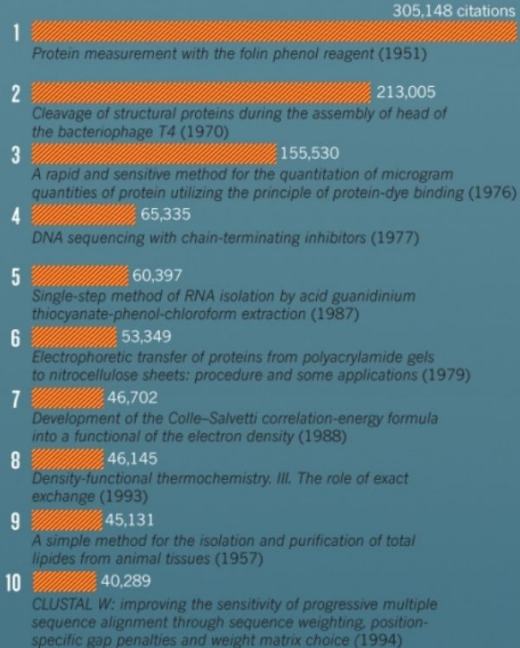




Methods Papers Are Highly Cited

TOP-10 PAPERS

Just 3 papers have received more than 100,000 citations, putting them well ahead of the rest. These runaway hits all cover biological lab techniques, which in general dominate the list of most-cited literature, including 7 of the top 10.

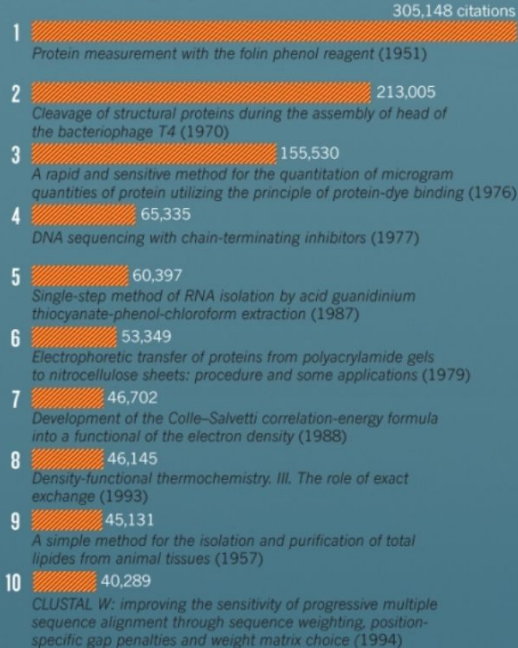




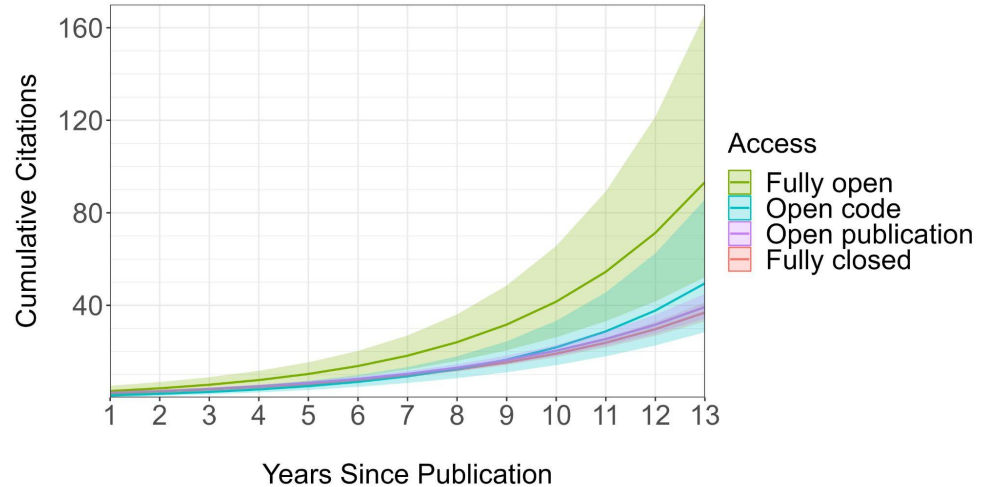
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(as are open access papers, so presumably Open Methods will be especially successful?)



What is Open Science?

Open Science: “transparent and accessible knowledge that is shared and developed through collaborative networks”



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Before next class:

- Prepare your presentations